PRO-ZERO CONES GIVE PRO-ISOMORPHISMS OF TRIANGLE SYSTEMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12749-12764, following the statement at 12737-12747.

Editorial note. No omissions: the five-term exact sequence is displayed.

Allowed standard prerequisites (editorial).

- Exactness of directed colimits. Algebra, Lemma algebra-lemma-directed-colimit-exact (cross-chapter, cited at ~12755). Colim preserves exactness of the Hom sequences.
- Pro-category recognition. Categories, Remark categories-remark-pro-category-copresheaves (cross-chapter, cited at ~12762). Converts the Hom-isomorphism statement into a pro-isomorphism.

Source reference. Original reference 00DB: focused-seeds/geometry/S9-der-pro-isomorphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0G2X: focused-seeds/geometry/S9-der-pro-isomorphism/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Let*

$$A_n \to B_n \to C_n \to A_n[1]$$

be an inverse system of distinguished triangles. If the system C_n is pro-zero (essentially constant with value 0), then the maps $A_n \to B_n$ determine a pro-isomorphism between the pro-object (A_n) and the pro-object (B_n).

Proof. For any object X of $\mathcal{D}$ consider the exact sequence

$$\operatorname{colim} \operatorname{Hom}(C_n, X) \to \operatorname{colim} \operatorname{Hom}(B_n, X) \to \operatorname{colim} \operatorname{Hom}(A_n, X) \to \operatorname{colim} \operatorname{Hom}(C_n[-1], X) \to$$

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Exactness follows from Lemma 0149 combined with Algebra, Lemma 00DB. By assumption the first and last term are zero. Hence the map $\operatorname{colim} \operatorname{Hom}(B_n, X) \to \operatorname{colim} \operatorname{Hom}(A_n, X)$ is an isomorphism for all X. The lemma follows from this and Categories, Remark 0G2X. $\square$